

Protective Collars with Standard Listed Options—TJUL as a Comparison

LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Construct a standard listed collar.** Match 100 shares with one ordinary listed put and one ordinary listed call using a common expiration.
- **Compute economic risk and reward.** Use executable premiums to calculate the floor, cap, breakeven, maximum loss, and maximum profit.
- **Compare implementation choices.** Distinguish a self-managed standard collar from TJUL's FLEX-option fund structure without requiring students to trade FLEX options.

1 The Core Position Uses Standard Listed Options

A protective collar combines long exposure to an underlying asset, a long put that limits downside, and a short call that helps finance the put by surrendering upside. The core classroom construction uses ordinary listed options that an options-approved brokerage account may support. FLEX options are not required.

Let one standard option contract represent $m = 100$ shares and hold n matched contracts. At time zero, buy or already own mn shares at S_0 , buy n puts with strike K_P , and sell n calls with strike K_C , where both option legs have the same expiration and

$$K_P < S_0 < K_C. \quad (1)$$

The terminal value is

$$V_T = mn [S_T + (K_P - S_T)^+ - (S_T - K_C)^+]. \quad (2)$$

Therefore,

$$V_T = mn \begin{cases} K_P, & S_T < K_P, \\ S_T, & K_P \leq S_T \leq K_C, \\ K_C, & S_T > K_C. \end{cases} \quad (3)$$

The put creates the floor, the shares participate one-for-one between the strikes, and the short call creates the cap.

The Options Clearing Corporation reports that each ordinary standard equity option generally covers 100 shares and that standard equity options are American-style. Corporate actions can create adjusted contracts with a different deliverable, so the broker's contract specification must still be checked.

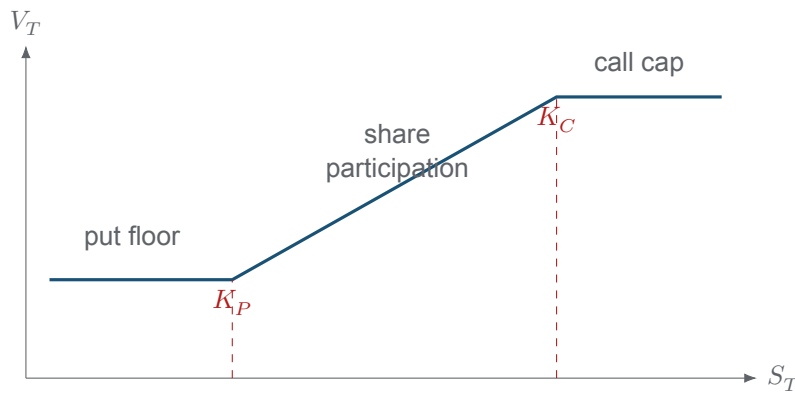


Figure 1: Terminal value of a standard listed protective collar. Premiums shift the profit diagram but do not change the three payoff regions.

2 Build the Retail Order Ticket

For an existing 100-share position, the opening option order is:

1. buy to open one put at strike K_P ; and
2. sell to open one call at strike K_C with the same expiration.

For a new position, buy 100 shares as well. Some brokers support a three-leg stock-plus-options combination order. Otherwise, entering the legs separately creates execution and interim-exposure risk.

Use executable quote sides in the pre-trade estimate. Let P_0^{ask} be the put ask and C_0^{bid} the call bid. A conservative initial cost for one matched collar is

$$\Pi_0 = m(S_0 + P_0^{ask} - C_0^{bid}), \quad m = 100. \quad (4)$$

The broker may fill a multi-leg limit order inside the component quotes, but a midpoint fill is an execution outcome rather than an assumption. Verify the underlying, deliverable, strikes, expiration, multiplier, option approval, buying power, and ex-dividend date before entering the order.

COMPANION NOTEBOOK

L11b: Standard-Listed SPY Protective Collar →

Match 100 illustrative SPY shares with one listed put and call, use executable quote sides, calculate risk metrics, and compare the result with TJUL.

3 Payoff Is Not Profit

Ignoring financing, dividends, taxes, and transaction fees, expiration profit is $V_T - \Pi_0$. For a matched collar,

$$\text{maximum loss} = \Pi_0 - mnK_P, \quad (5)$$

$$\text{maximum profit} = mnK_C - \Pi_0, \quad (6)$$

$$\text{interior breakeven} = \frac{\Pi_0}{mn} = S_0 + P_0^{ask} - C_0^{bid}. \quad (7)$$

The breakeven formula applies when it lies between the strikes. A “zero-cost collar” only means that the quoted put premium is approximately financed by the quoted call premium. It does not remove underlying risk, bid-ask cost, assignment risk, taxes, financing, or foregone upside.

4 The Standard Chain Constrains the Design

Standard listed options restrict the investor to available strikes and expirations. Raising K_P improves the floor but generally increases the put cost. Lowering K_C generally collects more call premium but surrenders upside sooner. A reproducible selection rule can maximize K_C subject to a minimum K_P , maximum net debit, minimum volume/open interest, and maximum spread.

For example, a 10%-down/10%-up collar has

$$K_P = 0.90S_0, \quad K_C = 1.10S_0. \quad (8)$$

It does not provide a 100% downside buffer. The position participates in the first 10% decline and is protected only below K_P . A put nearer S_0 provides stronger protection but normally requires a larger debit or a lower call cap.

5 Keep Growth and Price Changes Distinct

For a horizon Δt measured in years, this course uses

$$r_{t,\Delta t} := \log\left(\frac{S_{t+\Delta t}}{S_t}\right) = (\Delta t)g_{t,\Delta t}. \quad (9)$$

Option diagrams and defined-outcome documents generally quote a cumulative simple price change. Denote it by a capital R :

$$R_{t,\Delta t} := \frac{S_{t+\Delta t} - S_t}{S_t} = \exp((\Delta t)g_{t,\Delta t}) - 1. \quad (10)$$

A 90-day collar outcome or a two-year ETF cap is cumulative over its stated horizon, not an annual growth rate.

6 Expiration and Early Assignment

Below K_P , put exercise or assignment can sell the matched shares at the floor strike. Between the strikes, both options normally expire without exercise and the investor retains the shares. Above K_C , call assignment sells the shares at the cap strike.

Standard equity and ETF options are generally American-style, so assignment can occur before expiration. Short-call assignment risk is especially relevant near an ex-dividend date when remaining extrinsic value is small. The operator should decide before entry how to handle dividends, closing, rolling, exercise, assignment, and any mismatch between shares and adjusted deliverables. Adjusting one leg must not accidentally leave an unintended naked option.

7 TJUL Is the Comparison, Not the Required Instrument

The Innovator Equity Defined Protection ETF (TJUL) packages a defined outcome inside an exchange-traded fund. TJUL uses customized FLEX options on SPY; the disclosed construction uses purchased call exposure, a purchased put, and a sold call that creates the cap. FLEX options permit customized strikes, expirations, and exercise styles and use specialized exchange execution workflows. Students do not need that access. They can:

- build a self-managed collar with ordinary listed SPY options;

- buy or analyze TJUL shares as an ETF product; or
- compare the two without trying to replicate TJUL's FLEX positions.

Let R_{SPY} be the complete-period cumulative SPY price change and c the starting cap. A simplified full-buffer outcome before expenses is

$$R_{TJUL}^{\text{gross}} = \begin{cases} 0, & R_{SPY} < 0, \\ R_{SPY}, & 0 \leq R_{SPY} \leq c, \\ c, & R_{SPY} > c. \end{cases} \quad (11)$$

This is a complete-period outcome map, not a daily guarantee, a mid-period NAV model, or an instruction to trade FLEX contracts.

8 A Dated TJUL Snapshot

The official materials checked on August 4, 2026 describe an outcome period from July 1, 2025 through June 30, 2027, a 13.61% starting cap before expenses, a 100% starting buffer before expenses, and a 0.79% annual management fee. The prospectus illustrates a 12.03% cap and 98.42% buffer after two years of management fees. Refresh the [official fund page](#), [prospectus](#), and [factsheet](#) before class. Starting outcomes apply only from the first through the last day of the outcome period.

Feature	Self-managed standard collar	TJUL
Investor accesses	Shares plus listed put and call	ETF shares
Options used	Standard listed contracts	FLEX contracts managed inside fund
Granularity	Usually 100-share blocks	Whole or fractional ETF shares, broker dependent
Floor and cap	Available strikes, premium, expiration	Dated starting buffer and cap
Management	Investor handles execution and assignment	Fund manages option package
Costs	Spreads, fees, financing, taxes	Expense ratio, ETF spread, premium/discount, taxes

TJUL references the SPY *price* return. Direct SPY ownership may also receive distributions, so compare complete wealth using consistent price-return or total-return conventions.

9 Optional Advanced Track

An advanced standard-options project can load a dated SPY chain and optimize collar strikes under floor, cap, liquidity, and maximum-debit constraints. The previous Wheel lecture and supporting examples remain in the [advanced/](#) subfolder as a separate optional extension.

KEY TAKEAWAYS

In this lecture, we constructed a retail-replicable collar using ordinary listed options and used TJUL only as a defined-outcome comparison.

- **FLEX options are not required.** The core position is 100 shares, one standard listed put, and one standard listed call with a common expiration.
- **Use executable premiums.** Put ask minus call bid determines a conservative initial debit and therefore shifts breakeven, maximum loss, and maximum profit.
- **Implementation determines the outcome.** Listed strikes constrain a self-managed floor and cap; TJUL instead manages FLEX options internally while the investor accesses ETF shares.

Next, generalize sequential decisions to exploration–exploitation problems with stochastic multi-armed bandits.